Lok Jagruti Kendra University

Ahmedabad – 382210

Syllabus for Five Years Master of Science (Information Technology) Integrated Programme

Semester 7

(Effective from Academic 2022)

LOK JAGRUTI KENDRAUNIVERSITY

MSc.IT., Sem VII

Course Code	**050020701**			
Category	**Core Subject**			
Course Title	**Analysis and Design of Algorithms**			
Scheme and Credits	**Theory**	**Tutorial**	**Lab**	**Credits**
	3	**0**	**4**	**5**
Pre-requisites (if any)	Data Structure, C Programming Language.			

1. Course Objectives:

1.	Analyse algorithms by working out for time and space complexity of algorithms, design algorithm and growth of functions.
2.	Identify algorithm design methodology to solve problems. Analyse divide-and- conquer, probabilistic analysis and randomized algorithms.
3.	Able to understand dynamic programming, its elements and analyse existing algorithm of dynamic programming.
4.	To understand concept of greedy algorithms, amortized analysis and time complexity.
5.	Apply knowledge of minimum spanning trees and design algorithm for shortest paths

2. Course Contents:

Unit	Course Content	Weightage
I	**Basic concepts of Algorithms and its Analysis:** Algorithms, Algorithms as a technology, analysing algorithms(Space complexity and time complexity) , Designing algorithms, Asymptotic notations(Best, average and worst case), Standard notations and common functions for algorithms, Introduction to various methods of solving algorithms.	**15%**
II	**Algorithms Using Divide-and-Conquer** Recurrence relation, methods to solve recurrences (Master method, recurrence tree method and substitution method, Understanding of divide and conquer method using various algorithms like Stassen's matrix multiplication algorithm, Algorithm to multiply two large numbers, Binary Search, Sorting(Quick sort and merge sort), Maximum sub array problem.	**25%**
III	**Algorithm solving using Dynamic Programming:** Introduction to dynamic programming and its method, coin change problem, Rod cutting problem, Matrix-chain multiplication, Longest common subsequence, Optimal binary search trees.	**20%**
IV	**Greedy Algorithms** Introduction to Greedy algorithms and its analysis, knapsack problem, job sequencing with deadlines,	**20%**

	Huffman code algorithm, activity selection problem, Minimum spanning trees(Prims's and Kruskal's algorithm) Graphs shortest path algorithms Dijistra's and Bellman Ford Algorithms	
V	**Algorithms using Back Tracking and Branch and Bound method-** Introduction to backtracking and branch and bound method, n queens problem, 8 puzzle problem, knapsack problem using branch and bound, Depth First Search and Breadth first search algorithms.	**20%**

Desirable:

1. ICT enabled Classroom teaching
2. Case study
3. Practical / live assignment
4. Interactive class room discussions

3. Text Books:

1) Thomas H. Cormen, Charles E. Leiserson, Ronald L Rivest, Clifford Stein, Introduction to Algorithms, Third Edition, MIT Press (2009).
2) Parag H Dave, Himanshu B Dave, Design and Analysis of Algorithms Pearson (2014)
3) Thomas H. Cormen, Charles E. Leiserson, Ronald L Rivest, Clifford Stein Introduction to Algorithms, 2nd Edition PHI
4) Ellis Horowitz, Sartaj Sahni, Sanguthevar Rajasekaran Fundamentals of Computer Algorithms Universities Press (2008)
5) Gerard Tel Introduction to Distributed Algorithms Cambridge University Press (2004)

4. Accomplishment of the student after completing the course:

1) **Students will be able to identify the complexity of the problems and apply the method which is suitable to get the efficient result.**
2) **Students will be able to design better algorithms compared to the existing ones in terms of efficiency.**

LOK JAGRUTI KENDRAUNIVERSITY

MSc.IT., Sem VII

Course Code	**050020702**			
Category	**Core Subject**			
Course Title	**Advanced Java**			
Scheme and Credits	**Theory**	**Tutorial**	**Lab**	**Credits**
	0	**1**	**6**	**4**
Pre-requisites (if any)	**Programming Language of JAVA, HTML and JavaScript.**			

1. Course Objectives:

1.	To interpret the concepts of Servlet basics, Listener and Filers and easily use these concepts in Java technology.
2.	To develop proficiency in creating web based robust applications using the Java Server Pages (JSP).
3.	To understand concepts of Model View Controller (MVC), Java Database Connectivity (JDBC) applications and to implement them.

2. Course Contents:

Unit	Course Content	Weightage
I	**Servlet API and Overview** Servlet Basics, Basic Servlet structure, Servlets Generating text/html content, Packaging Servlets, The servlet life-cycle. Handling Client Request Form Data, Reading Form Data from Servlets, Handling Client Request, Reading Request Headers, Understanding HTTP/1.1 Request Headers, Changing the page according to how the user got there, Accessing the Standard CGI Variables. Generating the Server Response, HTTP Status Codes, Specifying Status Codes, HTTP / 1.1 Status Codes, Using Redirections, HTTP Response Headers, Setting Response Headers from Servlets, Understanding HTTP / 1.1 Response Headers. Handling Cookies, Remembering Usernames and Passwords, Deleting Cookies, Sending and Receiving Cookies, Using Cookie Attributes, Differentiating Session Cookies from Persistent Cookies, Using Cookies to Remember User Preferences, Session Tracking, Need for Session Tracking, Session Tracking API, Encoding URLs Sent to the Client, Accumulating a List of User Data.	**30%**
II	**Listeners and Filters** Using Servlet Context Listener, Http Session Listener, understanding of all the other Listeners viz. Servlet Request Listener, Servlet Context Attribute Listener, Servlet Request Attribute Listener, Http Session Attribute Listener	**15%**

	Using Filters for pre and post processing of request.	
III	**Java Server Pages , JSTL** JSP Basic Syntax, HTML Text, HTML comments, Template Text, JSP Comment, JSP Expression, JSP Scriptlet, JSP Declaration, JSP Directives, JSP Action, JSP Expression Language Element, Custom Tag (Custom Action), Escaped Template Text, Using JSP Scripting Elements, Using Predefined Variables, XML syntax for Expressions, Scriptlets, Declarations and Directives, Using Scriptlets, Using Declarations, Using Page Directive, Using Standard Actions Tags – <jsp:plugin>, <jsp:forward>, <jsp:include>, Using JavaBeans in JSP pages – <jsp:useBean>, <jsp:getProperty>, <jsp:setProperty>, Sharing Beans, Use of Scopes and their Attributes. Tag Library – Basics; Using JSTL – c:out, c:forEach, c:forTokens, c:if, c:choose, c:set, c:remove, c:import, c:url, c:param, c:redirect and c:catch Tags	**15%**
IV	**Model View Controller (MVC), EL** Integrating Servlets and JSP in a Web Application (MVC Architecture for Web Applications), Implementing MVC with Request Dispatcher, Understanding Data Sharing Between Servlets and JSP, JSP Expression Language, Accessing Scoped Variables, Bean Properties, Collections and Implicit Objects Using EL, Using EL Operators.	**20%**
V	**Java Web Frameworks:** Spring MVC Overview of Spring, Spring Architecture, bean life cycle, XML Configuration on Spring, Aspect – oriented Spring, Managing Database, Managing Transaction	**20%**

Desirable:

a) Hibernate 4.0 Overview of Hibernate, Hibernate Architecture, Hibernate Mapping Types, Hibernate O/R Mapping, Hibernate Annotation, Hibernate Query Language

a. Book: Hibernate 2nd edition, Jeff Linwood and Dave Minter, Beginning Après publication (Chapter 1,2,3)

b) Java Server Faces2.0

c) Servlet using JDBC

d) Using Servlets to Generate JPEG Images

3. **Text Books:**

1. Herbert Schildt, Java™: The Complete Reference, Seventh Edition

2. Marty Hall, Larry Brown, “Core Servlets and Java Server Pages Volume – 1: Core Technologies”, Pearson Education, 2nd Edition.

3. Marty Hall, Larry Brown, “Core Servlets and Java Server Pages Volume – 2: Advanced Technologies”, Pearson Education, 2nd Edition.

4. Spring in Action 3rd edition, Craig walls, Manning Publication

4. Webliography:

- http://docs.oracle.com/javaee/6/tutorial/doc/bnafd.html

5. Accomplishment of the student after completing the course:

- The student should be comfortable with developing web applications using Java technology.

Course Code	**050020703**			
Category	**Core Subject**			
Course Title	**Machine Learning using Python**			
Scheme and Credits	**Theory**	**Tutorial**	**Lab**	**Credits**
	3	**0**	**4**	**5**
Pre-requisites (if any)	Basics of computer science including algorithms, data structure, Basic Linear algebra and Probability theory.			

1. Course Objectives:

1.	Identify problems that are amenable to solution by ML methods, and which ML methods may be suited to solving a given problem.
2.	Analyze a given problem in the language/framework of different ML methods
3.	Implement basic ML algorithms.
4.	Design and carry out an empirical evaluation of different algorithms on problem formalization, and state the conclusions that the evaluation supports.

2. Course Contents:

Unit	Course Content	Weightage
I	**Introduction** Introduction to Machine Learning: What is human learning? What is Machine Learning? Human learning versus machine learning, Types of machine learning, Applications of machine learning, Tools for machine learning	**25%**
II	**Preparing to Model , Feature Engineering:** Machine Learning activities, Basic Types of data in Machine Learning, Structures of data, Data Quality and Remediation, Data Pre-Processing, Introduction to Feature Engineering, Feature Transformation, Feature Subset Selection	**20%**
III	**Modelling and Evaluation:** Selecting a Model, Training a Model, Model Representation and Interpretability, Evaluating Performance of a Model, Improving Performance of a Model	**20%**
IV	**Supervised Learning:** Classification and Regression: Example of Supervised Learning, Classification Model, Classification Learning Steps, Common Classification Algorithms Example of Regression, Common regression Algorithms	**20%**

V	**Unsupervised Learning:** Unsupervised Learning versus Supervised Learning, Applications of Unsupervised Learning, Clustering, Finding Pattern using Association Rules	**15%**

3. Text Books:

- Applied Machine Learning, M. Gopal, McGraw Hill Education
- Machine Learning: A Probabilistic Perspective, Kevin Murphy, MIT Press,2012
- The Elements of Statistical Learning, Trevor Hastie, Robert Tibshirani, Jerome Friedman, Springer 2009 (freely available online)
- Pattern Recognition and Machine Learning, Christopher Bishop, Springer,2007
- Programming Collective Intelligence: Building Smart Web 2.0 Applications - Toby Segaran
- Building Machine Learning Systems with Python - WilliRichert, Luis Pedro Coelho
- Machine Learning Saikat Dutt, Subramanian Chandramouli, Amit Kumar Das - Pearson Education

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MSc.IT., Sem VII

Course Code	**050029704**			
Category	**Elective Subject**			
Course Title	**Software Testing**			
Scheme and Credits	**Theory**	**Tutorial**	**Lab**	**Credits**
	4	**0**	**2**	**5**
Pre-requisites (if any)	1. **Basic knowledge of Java Programming** 2. **Software Engineering concepts**			

1. Course Objectives:

1.	To understand the concept and importance of software testing
2.	To have the knowledge of different testing techniques
3.	To differentiate between manual testing and automated testing
4.	To have a clear knowledge of testing documentation

2. Course Contents:

Unit	Chapter Details	Weightage
I	**Introduction to testing:** What is Software testing, What is Software Quality, Importance of testing, Principles of Software Testing, Software Testing Terminology, Verification and Validation, Software Quality Assurance vs Software Quality vs Software Testing, Different roles of testing, SDLC and STLC, STLC Phases: Requirement Analysis, Test Planning, Test Design, Test Execution, Test Scenario Preparation, Test Case Preparation, Test Environment and Test Data Preparation, Requirement Traceability Matrix, Software Testing Methodology	**25%**
II	**Introduction to SDLC and STLC:** Overview on SDLC, Different Life Cycle models: Waterfall, V model, Prototype model, Spiral model, Iterative model and SCRUM **Software Testing as per development methodology:** TDD, Kanban, DevOps, Overview on STLC and AGILE Testing	**10%**

III	**Testing techniques & Levels of Testing:** Manual and Automated testing, Dynamic Testing (Black Box Testing), Static Testing (White Box Testing), Grey box testing, Functional and Non-functional testing, **Levels of testing:** Unit Testing, Integration Testing, System Testing, User Acceptance Testing **Types of testing (Others):** Positive Testing, Negative Testing, Beta Testing, Formal Testing, Informal Testing, Monkey Testing, Re-Testing **Special Tests:** Security Testing, Performance Testing, Installation Testing, Regression Testing, Smoke Testing, Sanity Testing, Adhoc Testing, Usability Testing, COTS Testing etc.	**25%**
IV	**Test Planning, Test Metrics and Reports** **i) Test Planning:** Test policy, Test Strategy, Test plan, Quality Plan, Test Estimation, Test Standards, Build Test data and Test cases, Test Scenarios, Essential activities in testing, Test Script, Template for test cases, Tools used to build test data, Roles and Responsibilities in testing life cycle, Test progress monitoring **ii) Test Metrics:** Testing related data, Defect data, Efficiency/Productivity data, Estimated, Budgeted, Approved and Actual, Defect density, Residual Defect density, MTBF/MTTR **iii) Test Reports** Test Report, Project test status report, Integration Test Report, System test Report, Acceptance Test Report, Guidelines for Test writing and using test report, Test Status report	**25%**
V	**Testing and Testing management tools:** Features of testing tools, Guidelines for selecting a tool, Advantages and disadvantages of using tools · Dynamic and Static testing tools · Testing using Automated testing tools · Process of procurement of COTS (Readily available tools from Market)	**15%**

Seminars

TFS (Microsoft Team Foundation Server)

- Jira
- Mantis
- Open Source and Licensed tools (HP, IBM and others)

3. Text Book(s):

- M G Limaye, Software testing Principles, techniques and Tools, McGraw Hill
- Naresh Chauhan, Software testing Principles and Practices, Oxford University Press
- Renu Rajani, Pradeep Oak, "Software Testing – Effective Methods, Tools and Techniques", Tata McGraw Hill, Latest Edition.

- Nina S Godbole, Software Quality Assuarance – Principles and Practice, Narosa Publising house.
- Srinivasan Desikan and Gopalaswamy Ramesh, “Software Testing – Principles and Practices”, Pearson education
- Aditya P.Mathur, “Foundations of Software Testing”, Pearson Education
- Boris Beizer, “Software Testing Techniques”, Second Edition, Dreamtech
- Elfriede Dustin, “Effective Software Testing”, First Edition, Pearson Education

4. Webliography:

- **https://www.coursera.org/specializations/software-testing-automation#courses**
- **https://www.guru99.com/software-testing.html**
- **https://www.tutorialspoint.com/jira/index.htm**
- **https://www.guru99.com/bugzilla-tutorial-for-beginners.html**
- **https://www.selenium.dev/**
- **https://www.javatpoint.com/selenium-tutorial**

5. Accomplishments of the student after completing the course:

- Students will understand software testing process and execution of software testing, different types of testing techniques, will be clear about project and domain specific knowledge of both Manual Testing techniques and Automated testing.

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MSc.IT., Sem VII

Course Code	**050029705**			
Category	**Elective Subject**			
Course Title	**Cyber Security**			
Scheme and Credits	**Theory**	**Tutorial**	**Lab**	**Credits**
	4	**0**	**2**	**5**
Pre-requisites (if any)	Knowledge of networks and networking technologies			

1. Course Objectives:

Sr.	Course Outcome (Learner will be able to)
1.	Understand the fundamental concepts of cyber security and apply them in practical scenarios.
2.	Identify various types of cyber threats and attacks, and evaluate defence mechanisms.
3.	Implement secure coding practices to prevent and mitigate vulnerabilities.
4.	Design secure networks and understand network security best practices.
5.	Apply ethical hacking techniques to test system vulnerabilities.

2. Course contents:

Module	Content	Weightage
Unit I	**Introduction to Cyber Security** Definition of Cyber Security, Importance of Cyber Security, Principles of Cyber Security, Fundamentals of Information Security: CIA Triad (Confidentiality, Integrity, Availability), Role of Cybersecurity in Modern Organizations, Risk Analysis and Security Frameworks Basics of risk analysis and management: Identify Risks, Risk Assessment, Prioritize Risks, Risk Mitigation Strategies	15%

Unit II	**Cyber Threats and Attacks** Types of cyber threats: active and passive attacks. **Malware Attacks:** Viruses, Worms, Trojans, Ransomware, Spyware, Adware. **Phishing Attacks:** Email phishing, Spear phishing, Vishing (voice phishing), Smishing (SMS phishing). **Denial-of-Service (DoS) Attacks:** Distributed Denial-of-Service (DDoS) attacks, SYN flood attacks, UDP flood attacks. **Man-in-the-Middle (MitM) Attacks:** Session hijacking, Wi-Fi eavesdropping. **Insider Threats:** Data theft (financial, reputational, legal), Unauthorized access, Social engineering by insiders. **IoT Attacks**: Botnet attacks, Device hijacking, Firmware attacks. Zero-Day Exploits, Social Engineering, sybli attack	25%

Unit III	**Network Security** Network Fundamentals: Basics of networking and TCP/IP protocol suite, Common network vulnerabilities, Firewalls and types, Intrusion Detection Systems, Intrusion Prevention Systems and types, Virtual private network (VPNs) and types, proxy servers and types, secure remote access and network segmentation. Capture the network using Wireshark, Installation VPN and Proxy server. **Application Security** Secure Coding Practices, Common Vulnerabilities (e.g., OWASP Top Ten), Web Security, and Database Security, Password cracking techniques, Key loggers, Steganography, Owasp top 10 checking vulnerabilities using DVWA (Damn Vulnerable Web Application) and burp suite.	MID 25%
Unit IV	**Cryptography and Cryptographic Techniques** Fundamentals of Cryptography: Definition and objectives of cryptography, Importance of cryptographic, Components of a cryptographic system, Cryptographic primitives: encryption, decryption, hashing, Symmetric Cryptography, Principles of symmetric encryption, Asymmetric Cryptography, Principles of asymmetric encryption, Hash Functions and Message Authentication Codes (MAC), Authentication protocol: Kerberos, digital signatures and public key infrastructure (PKI).	25%
Unit V	**Cybercrime and Cyber law** Reporting of cybercrimes, Remedial and mitigation measures, Legal perspective of cybercrime, IT Act 2000 and its amendments, Cybercrime and offences, Organisations dealing with Cybercrime and Cyber security in India, Case studies **Tools and Utilities: Using Basic Tools** IFconfig/IPconfig, Whois, Nslookup, PING, Traceroute, Telnet, Secure Shell, Monitoring Tools and Software: Nagios, SolarWinds, Microsoft Network Monitor, Wireshark, Snort, Nmap, Nikto, OpenVAS, Metasploit, The Browser Exploitation Framework	10%

3. Desirable:

1) Familiarity with Linux operating system.

2) Hands-on labs or projects focusing on practical applications of cyber security concepts.

4. Main Text Books:

1) Principles of Computer Security" by W. Stallings and L. Brown.

2) Cryptography and Network Security: Principles and Practice" by W. Stallings

5. Webliography:

1. https://www.tutorialspoint.com/information_security_cyber_law/quick_guide.htm
2. https://www.javatpoint.com/cyber-security-tutorial
3. https://www.slideserve.com/promisec/how-vulnerability-scanning-works

5. Accomplishment of the student after completing the course:

- After completion of this course, students can find and fix the vulnerabilities in system, network and web application and also have knowledge of cyber-crime and cyber laws.

Course Code	**050029706**			
Category	**Elective Subject**			
Course Title	**Fundamentals of Digital Marketing**			
Scheme and Credits	**Theory**	**Tutorial**	**Lab**	**Credits**
	4	**0**	**2**	**5**
Pre-requisites (if any)	NIL			

1. **Course Objectives:**

1.	To understand the basics of Digital Marketing in today's competitive world. Learn the difference between Traditional Marketing and Digital Marketing. Create digital strategy development and planning

2. **Course Contents: -**

Unit	Topic	Weightage
I	**Introduction of Marketing & Digital Marketing: -** Understanding traditional marketing. Learn 5 Ps of marketing. TCEO Model (Think, Create, Engage & Optimize). Understanding of Digital Marketing, Types of Digital Marketing & Channels. Benefits and Advantages of Digital Marketing.	**20%**
II	**Digital Marketing Methodologies I** **Search Engine Advertising:** Pay for Search Advertisements, Ad Placement, Ad Ranks, Creating and Enhancing Ad Campaigns, Evaluating Campaigns **Display marketing:** Display Ads – Concepts and Types, - Buying Models (CPC, CPM, CPL, CPA), Targeting Display Ads, - Programmable Digital Marketing, - Analytical Tools **YouTube marketing**	**20%**
III	**Social Media Marketing** Why Social Marketing over Traditional Marketing. Social Media vs Digital Media, Introduction to social media platforms, penetration & characteristics; Building a successful social media marketing strategy **Facebook Marketing:** Business through Facebook Marketing, Creating	**30%**

	Advertising Campaigns, Adverts, Facebook Marketing Tools & Insights **Linked in Marketing:** Introduction and Importance of Linked in Marketing, Framing Linked in Strategy, Lead Generation through Linked in, Content Strategy, Analytics and Targeting **Twitter Marketing: -** Introduction to Twitter Marketing, how twitter Marketing is different than other forms of digital marketing, framing content strategy, Twitter Advertising Campaigns **Instagram and Snapchat:** Digital Marketing Strategies through Instagram and Snapchat **Introduction to social media metrics**	
IV	**Mobile Marketing and Web Analytics** **Mobile Marketing:** Mobile Advertising, Forms of Mobile Marketing, Features, Mobile Campaign Development, Mobile Advertising Analytics Web Analytics: - Google Analytics & Google AdWords; data collection for web analytics, multichannel attribution, Universal analytics, Tracking code	**20%**
V	**Introduction to SEO, SEM** Introduction to SEO, SEM: Introduction and need for SEO, how to use internet & search engines; search engine and its working pattern, On-page and off-page optimization, SEO Tactics - Introduction to SEM	**10%**
VI	A group of three students (Maximum) has to work on creating an advertising campaign through any form of digital marketing viz: Mobile Marketing, Twitter Marketing, Facebook Marketing, LinkedIn Marketing, Instagram or Snapchat Marketing. The student/s should work on creating the campaign, running the campaign, presenting the results of the campaign in terms of Lead Generation and / or sales and / or web analytics	**CEC**

Desirable:

- Workshop on topic listed in Unit II, III and IV (student driven).

3. **Text Books:**

1. Swaminathan T.N. / Karthik Kumar, Digital Marketing – From Fundamentals to Future
2. Rob Stokes and the Minds of Quirk ,E-Marketing – The Essential Guide to Digital Marketing (5th Edition)
3. Seema Gupta, Digital Marketing, Mc-Graw Hill
4. Puneet Singh Bhatia, Fundamentals of Digital Marketing, Pearson
5. Ian Dodson, The Art of Digital Marketing, Wiley
6. Social Media Marketing: A Strategic Approach By -Melissa S. Barker | Donald I. Barker | Nicholas F. Bormann | Debra Zahay | Mary Lou Roberts Cengage
7. Digital Marketing By - Dr. Ragavendra K. and Shruthi P. Himalaya Publication
8. Digital Social Media Marketing By -Prof. Nitin C. Kamat, Chinmay Nitin Kamat Himalaya Publication

4. Webliography:

- https://learndigital.withgoogle.com/digitalunlocked/
- https://digitalskills.fb.com/en-in/
- https://www.hubspot.com/digital-marketing

5. Accomplishment of the student after completing the course:

- After the completion of the course, the student will gain basic knowledge on how to use the Social Media for marketing. Learn the concepts of SEO, Web Analytics, Influencers marketing & Content marketing. The student will get a brief overview on LinkedIn Marketing, Instagram Marketing, Facebook Marketing, Email Marketing and You Tube Marketing.

Course Code	**050020708**			
Category	**Software Project Using FULL STACK (MERN)**			
Course Title	**Project**			
Scheme and Credits	**Theory**	**Tutorial**	**Lab**	**Credits**
	0	**1**	**8**	**5**
Pre-requisites (if any)				

Guidelines

- The project definition should be finalized during the First week of Semester start Any 'good' internal definition having a high application potential will also be acceptable
- It is recommended that the team should be about 2-3 students, but not more than 4 students.
- Project plan along with the division of work amongst teammates would have been prepared and got approved within a maximum of 5 days of the start of the project.
- Coding standards should be followed meticulously. At the minimum, the code should be self-documented, modular, and should use the meaningful naming convention.
- It is advisable that object-oriented methodology is used with the reusability of classes and code, etc.
- The output reports must include MIS reports, if applicable.
- The documentation should include a chapter on "Learning during Project Work", i.e. "Experience of Journey during Project Duration".
- Data structure (database design) is mandatory. At least portions of code (preferably full code) are mandatory. The student may be asked to write the code related to the project during the examination.
- If a student is compelled to follow certain instructions (by the external, i.e. organization's Guide) which he/she does not agree to, such a student must prepare a supplementary report to document his/her version and present it to the examiners if such a need arises
- Internal guides (i.e. the faculty members) must devote the time allocated as per the timetable to guide the students for the project. The time allocation will be in accordance with the scheme for the 6th semester project as given.
- Project document should be printed on both sides of paper.

Documentation:

- The project has to be well-documented in the form of a Project Report (at least 50 pages comprising of the design, data dictionary, source code, screen-shots, etc.).
- Format: Print out on both the side of page with single line (preferably one-and-ahalf line) spacing. Use Times New Roman of size 12 for normal text.
- **TABLE OF CONTENTS**
    - 1. Introduction
        - 1.1. Existing System

        - 1.2. Need for the New System
        - 1.3. Objective of the New System
        - 1.4. Problem Definition
        - 1.5. Scope of the Project and Core Components
        - 1.6. Project Profile
        - 1.7. Assumptions and Constraints
        - 1.8. Advantages and Limitations of the Proposed System

    - 2. Requirement Determination & Analysis
        - 2.1. Requirement Determination
        - 2.2. Targeted Users
    - 3. System Design
        - 3.1. Use Case Diagram
        - 3.2. Class Diagram
        - 3.3. Interaction Diagram
        - 3.4. Activity Diagram
        - 3.5. Data Dictionary
    - 4. Development
        - 4.1. Coding Standards
    - 5. Agile Documentation
        - 5.1 Agile Project Charter
        - 5.2 Agile Road map / Schedule
        - 5.3 Agile Project Plan
        - 5.4 Agile User Story (Minimum 3 Tasks)
        - 5.5 Agile Release Plan
        - 5.6 Agile Sprint Backlog
        - 5.7 Agile Test Plan
        - 5.8 Earned-value and burn charts
    - 6. Proposed Enhancements
    - 7. Conclusion
    - 8. Bibliography 12

- **Evaluation Parameters :**

    - Evaluation of the projects would be done considering the framework available at the Institute.
    - The main parameter of assessment would be the ability of the students to code.
    - Though the project and domain specific knowledge would be assessed for, the evaluation would predominantly depend on the students' ability to explain, modify or revise of code.
    - Coding standards should have been implemented.

- Though the project would be evaluated for the entire team, the examiner should emphasize on the contribution of each team member in the project development
- Total Marks ((%age provided in below table)

Explanation of Code	50%
Documentation - Agile	20%
Explanation of Analysis and Design	20%
Presentation	10%